Seamless Global AOD Reconstruction Based on Transformer Model and Fusion of Multisource Satellite Products

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***Abstract*—Aerosol optical depth (AOD), as a key parameter reflecting the atmospheric aerosol load, plays a crucial role in the study of climate change, air quality, and the Earth's radiation budget. However, current satellite-retrieved AOD products suffer from high missing rates, spatial discontinuities, and interproduct inconsistencies, which significantly limit their global applications. In this study, we propose a seamless reconstruction framework based on a cross-dimensional self-attention (CDSA) transformer, by combining optimal interpolation (OI) fusion of multiple satellite products with transformer-based imputation to generate globally continuous AOD fields with high accuracy. First, eight AOD products from MODIS and Visible Infrared Imaging Radiometer Suite (VIIRS) sensors are optimally fused, which substantially improves both coverage and accuracy across the entire sample range. Subsequently, a transformer imputation model that integrates temporal, spatial, and meteorological variables is employed to seamlessly recover missing regions through cross-dimensional attention mechanisms, further enhancing data completeness and reliability. Experimental results indicate that: 1) compared with individual satellite products, the OI fusion significantly reduces errors and improves correlations with aerosol robotic network (AERONET) observations, doubling the overall coverage; 2) the transformer-based imputation achieves 100% coverage while further increasing the coefficient of determination (R^2) and reducing the root mean square error (RMSE), outperforming the reanalysis product MERRA-2; and 3) at the regional scale, the imputed results accurately capture the spatiotemporal evolution of typical pollution episodes. The final global product reaches an overall accuracy of $R^2 = 0.701$ and RMSE = 0.073. This work provides a feasible pathway for generating seamless global AOD datasets, offering essential support for climate model validation, air pollution monitoring, and environmental policy-making.**

***Index Terms*—Multisource product fusion, satellite aerosol optical depth (AOD) products, seamless global reconstruction, transformer model.**

I. Introduction

ATMOSPHERIC aerosols, as mixtures of suspended solid and liquid particles in the atmosphere, are central subjects in studies of the Earth's climate system, ecological environment, and human health. They directly scatter and absorb solar radiation and modify cloud microphysical properties. These effects exert profound impacts on the global energy balance, making them one of the largest sources of uncertainty in climate prediction [1], [2], [3], [4]. In addition, fine aerosol particles such as PM2.5 can penetrate the human respiratory system, increasing the risks of cardiovascular and respiratory diseases [5], [6], [7]. Environmental issues such as reduced visibility and acid rain formation are also closely associated with aerosol concentrations. Aerosol optical depth (AOD), defined as the vertical integral of the aerosol extinction coefficient within an atmospheric column, directly quantifies the total aerosol burden and has been widely employed to estimate radiative forcing, infer near-surface particle concentrations, drive air-quality models, and validate climate model parameterizations.

Satellite remote sensing has become the primary means of obtaining AOD due to its global coverage and long-term observation capabilities [8]. Currently, major international AOD products include those from polar-orbiting sensors such as MODIS (Terra/Aqua), Visible Infrared Imaging Radiometer Suite (VIIRS) (SNPP/NOAA20), and MISR (Terra), as well as geostationary sensors such as Himawari-8 AHI and GOES [9], [10], [11], [12]. These products provide critical data support for regional and global aerosol research.

Nevertheless, existing satellite AOD products face several significant limitations that hinder their utility in high-precision applications. Due to limited stable satellite coverage at higher latitudes and the influence of Earth projection distortion, this study focuses on data coverage within the latitudinal range from 60°S to 60°N. In subsequent discussions on coverage and missing rates, this region is accordingly defined as the global domain for analysis. First, data missing rates remain

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high due to factors such as cloud cover, high surface albedo (e.g., snow and desert), and large solar zenith angles, which lead to retrieval failures. Globally, missing rates commonly range from 30% to 70%, and exceed 80% in tropical rainy regions, high-latitude winters, and complex terrains such as the Tibetan Plateau [13], [14], [15]. For instance, during the Amazon rainy season, valid MODIS AOD retrievals account for less than 20% of observations, limiting the ability to capture seasonal aerosol variations [16]. Second, cross-product consistency remains poor. Differences in sensor observation geometry (e.g., scan angle, overpass time), retrieval algorithms [e.g., dark target (DT) and deep blue (DB)], and radiometric calibration lead to substantial biases in both accuracy and spatial distribution. For example, comparisons between MODIS Collection 6.1 and VIIRS SNPP products reveal deviations up to ± 0.3 in polluted East Asian regions and about ± 0.05 even in clean marine areas [17], [18], [19], posing challenges for multisource fusion and long-term trend analysis.

To address these challenges, extensive research has been devoted to AOD data fusion and imputation, forming two major methodological categories: multisource fusion and missing-value interpolation. Multisource fusion methods can be classified into single-pixel and multipixel approaches. Single-pixel methods establish statistical relationships between overlapping products, such as polynomial regression or optimal interpolation (OI), where higher weights are assigned to more reliable products based on error covariance matrices. These approaches are straightforward and can improve accuracy in overlapping regions (e.g., reducing AOD errors by ~15% when fusing MODIS and MISR), but they fail to address large "data gaps" where all products are missing, leading to discontinuous fused datasets [20], [21], [22]. In contrast, multipixel methods exploit spatial and temporal correlations to estimate missing regions, including geostatistical models such as universal kriging, Bayesian maximum entropy (BME), spatiotemporal statistical fusion (SSDF), and fixed-rank smoothing. For instance, BME has been shown to increase coverage from 50% to 85% in the Beijing–Tianjin–Hebei region. However, it struggles to handle heterogeneous product accuracy and redundancy; when the number of input products exceeds three, computational costs grow exponentially, limiting its applicability at the global scale [23], [24], [25]. Moreover, most fusion studies remain regional (e.g., East China and North America), with few addressing seamless global-scale solutions.

Interpolation approaches can also be categorized into three groups.

1) Statistical methods based on auxiliary variables, such as topography (DEM), meteorological parameters (wind speed and relative humidity), and land cover, can perform well in homogeneous regions but show significant accuracy declines (RMSE > 0.2) in complex landscapes such as urban–rural transitions [26].

2) Machine learning models such as random forests and support vector machines (SVMs) can capture nonlinear relationships between AOD and auxiliary factors, achieving RMSE = 0.150 in regions such as the Yangtze River Delta. However, their generalization ability degrades in data-sparse areas like the Sahel [27].

3) Deep learning approaches, including convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, have achieved high accuracy in local studies (e.g., RMSE = 0.120 in Beijing), but CNNs struggle with long-range dependencies, LSTMs have difficulty incorporating diverse auxiliary inputs, and both approaches face the common challenge of poor transferability from regional to global scales [28], [29], [30].

Since its introduction in the natural language processing (NLP) field in 2017, the transformer architecture has revolutionized the paradigm of sequence modeling with its fully self-attention-based mechanism [31]. Its core strength lies in the ability to directly compute dependencies between any two elements in a sequence, thereby efficiently capturing long-range contextual information while avoiding the gradient vanishing problem of recurrent neural networks (RNNs) and the limited receptive field of CNNs. This characteristic has enabled its rapid expansion into domains such as computer vision and speech recognition, where it has demonstrated exceptional performance in tasks including image generation and video understanding.

Compared to traditional models, the transformer is better equipped to capture complex relationships within data and overcomes several limitations associated with conventional approaches in handling time series. It has gradually been adopted in time-series forecasting tasks across various fields, such as atmospheric pollutant monitoring [68] and remote sensing-based water quality monitoring [69], demonstrating its strong capability in processing temporal data. By offering a new and effective methodology for time-series analysis and prediction, the transformer architecture has contributed to advancing development in these related research areas.

Current AOD imputation models face three major limitations: 1) insufficient integration of multidimensional correlations across time, space, and auxiliary variables; 2) limited global generalizability, as region-specific models often suffer an accuracy loss of more than 30% when applied across different latitudes or climate zones; and 3) underutilization of multisource satellite data, with most models relying on only one or two products, failing to leverage their complementarity [32], [33], [34]. These shortcomings have thus far prevented the generation of seamless, high-accuracy global AOD datasets.

To overcome these limitations, this study proposes a two-stage "multisource fusion + transformer-based imputation" framework. First, eight AOD products from MODIS and VIIRS are optimally fused to maximize the retention of valid observations and exploit complementary product strengths. Second, a novel transformer imputation model is developed to recover missing regions, incorporating a 3-D attention mechanism across temporal, spatial, and auxiliary domains. This framework, for the first time, enables the generation of seamless global daily AOD data at 0.1° resolution with high accuracy, providing critical datasets for atmospheric environment monitoring, climate model evaluation, and global change research.

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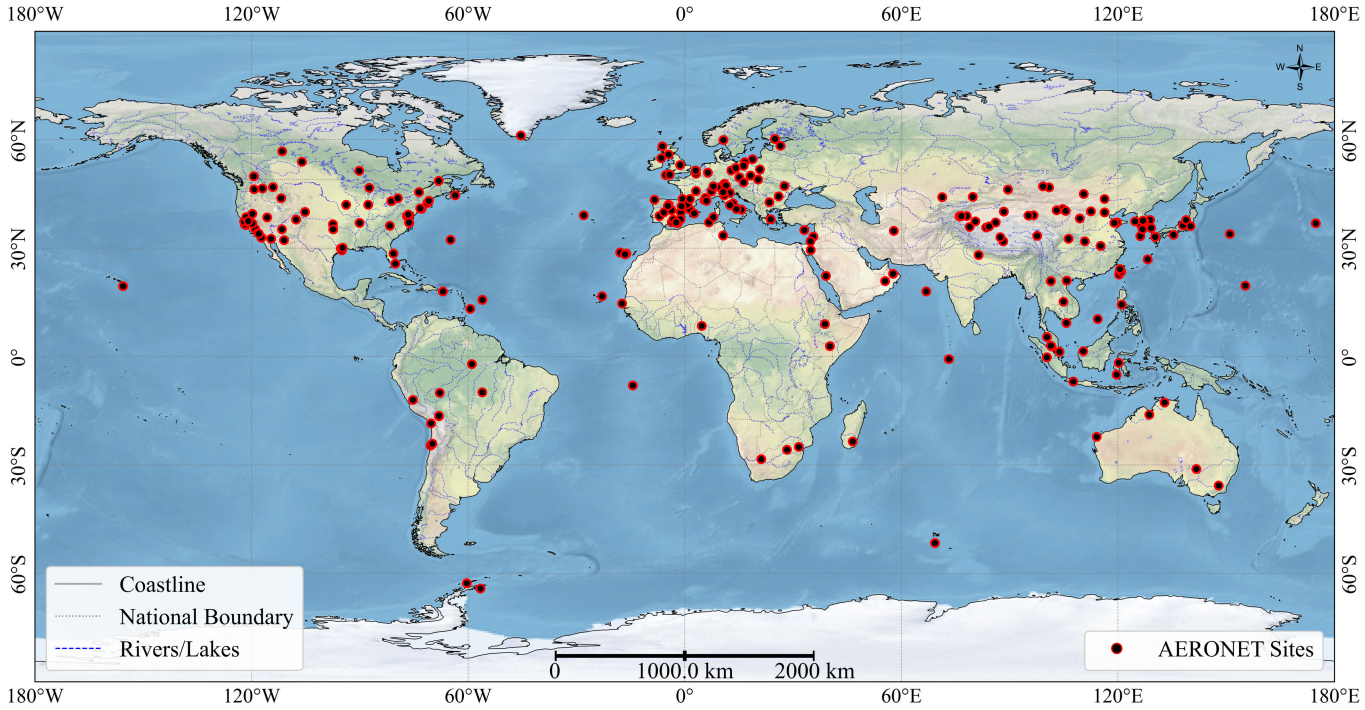


Fig. 1. Location of AERONET level 2.0 (red circles and black dots).

TABLE I
SUMMARY OF SATELLITE AOD PRODUCTS USED IN FUSION AND IMPUTATION

Platform	Sensor	Product	Quality Control	Spatial Resolution	Transit Time	Wavelength	Version
Aqua	MODIS	MYD04_3K DT	QA = 3	3km	13:30	550nm	Collection6.1
	MODIS	MYD04_L2 DT	QA = 3	10km	13:30	550nm	Collection6.1
	MODIS	MYD04_L2 DB	QA $\geqslant$ 2	10km	13:30	550nm	Collection6.1
	MODIS	MYD04_L2 DTB	-	10km	13:30	550nm	Collection6.1
SNPP	VIIRS	AERDT_L2_VIIRS_SNPP	QA = 3	6km	13:30	550nm	Version2.0
	VIIRS	AERDB_L2_VIIRS_SNPP	QA = 3	6km	13:30	550nm	Version2.0
NOAA20	VIIRS	AERDT_L2_VIIRS_NOAA20	QA = 3	6km	13:30	550nm	Version2.0
	VIIRS	AERDB_L2_VIIRS_NOAA20	QA = 3	6km	13:30	550nm	Version2.0

II. STUDY AREA AND DATASETS

A. Study Area

In previous studies, typical regions—such as a single city or a metropolitan agglomeration—were often selected as the study area to enable targeted analyses, thereby obtaining models and parameters most suitable for those regions and achieving higher local accuracy. However, such models generally perform well only within the selected region; their scalability and portability are limited, and the final results are confined to local applications.

In contrast, the continuity and completeness of AOD data are of vital importance for atmospheric environmental monitoring and studies of the Earth's radiation budget. At present, there is an urgent need to construct seamless global AOD remote sensing products. Therefore, as shown in Fig. 1, this study selects the global domain as the study area. Moreover, Fig. 1 also depicts the distribution of all aerosol robotic network (AERONET) Level 2.0 sites utilized for validation in the experiment.

B. Experimental Datasets

The original satellite AOD inputs used in this study comprise eight AOD products from the MODIS and VIIRS sensors. In addition, MERRA-2 and ERA5 reanalysis datasets are used as auxiliary data. AERONET ground sites are selected to validate and evaluate the accuracy of each AOD product.

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Table I summarizes the eight satellite AOD products used in the fusion and imputation experiments.

To ensure the reliability of the input data, strict quality control (QC) was applied to all satellite AOD products in this study, primarily based on the inherent quality assurance (QA) flag layer accompanying each dataset. The QA flag is an integer value that indicates the retrieval quality, cloud masking status, and surface suitability for each pixel, with higher values generally corresponding to greater confidence and lower errors.

For MODIS DT products, the official "high-quality" standard was adopted, i.e., QA = 3. This grade requires the retrieval to be of the best quality and applicable over land or ocean surfaces suitable for aerosol modeling [8]. MODIS DB products were filtered using QA ≥ 2, which includes retrievals of medium quality or higher, thereby retaining sufficient data volume while excluding questionable or invalid observations [35]. VIIRS products employed the QA = 3 threshold, corresponding to the high-confidence retrieval defined in the VIIRS aerosol algorithm, which has undergone rigorous cloud screening and surface-type validation [38]. As for the MODIS DTB (combined DT–DB) product, which automatically merges DT and DB results via the official algorithm, its QA flag represents the best merged estimate; thus, the default output was used without additional filtering.

During the data preprocessing stage, the following steps were applied to the downloaded original Level-2 products: first, the AOD data layer and its corresponding QA flag layer were read from each file. A binary quality mask was then generated based on the product-specific QA thresholds described above. Pixels satisfying the threshold criteria retained their AOD values, while those failing the criteria were marked as missing (NaN). This step was performed prior to data resampling and gridding, ensuring that only consistently quality-controlled, high-confidence observations were used in subsequent fusion and modeling.

1) MODIS Dataset: The Moderate Resolution Imaging Spectroradiometer (MODIS) aboard the Aqua satellite is one of the core sensors of NASA's Earth Observing System (EOS). Its aerosol products are characterized by high spatiotemporal coverage and multialgorithm synergy, providing key data support for global aerosol monitoring. Aqua was launched in 2002 into a sun-synchronous orbit at an altitude of 705 km; together with Terra, it forms a "morning–afternoon" dual-satellite observation system. The sun-synchronous design ensures that the same region is observed daily at fixed local times, reducing the impact of solar zenith angle variations on aerosol retrievals. Focusing on afternoon overpasses, this study uses four Aqua-MODIS Level-2 aerosol products for January 2024, all at 550 nm: MYD04_3K (3 km), MYD04_L2 DT (10 km, DT), MYD04_L2 DB (10 km, DB), and MYD04_L2 DTB (10 km, DT–DB combined). Aqua's local overpass time is about 13:30. Compared with Terra-MODIS, the Aqua afternoon overpass better captures aerosol distributions after the boundary layer fully develops—e.g., near peak pollution accumulation periods [35], [36], [37]. The four products jointly resolve both microscale spatial variability and macroscale distributions. With overpass times aligned to VIIRS on SNPP and NOAA20, the datasets are mutually complementary for cross-validation and fusion.

2) VIIRS Dataset: The VIIRS aboard SNPP and NOAA20 operates in sun-synchronous orbits with fixed local overpass time (13:30), providing daily global scans at high spatiotemporal resolution. Both satellites belong to the JPSS program at ~833-km altitude. As the successor to MODIS, VIIRS inherits multispectral observing capabilities; its 3000-km swath notably improves data continuity over mid- to low-latitudes and performs better over complex terrain and cloudy regions. This study uses NASA Level-2 aerosol products for January 2024, including the AERDT_L2 and AERDB_L2 series at 6 km resolution and 550 nm, Version 2.0 with strict QA screening. AERDT_L2 implements an improved DT algorithm with separate land and ocean modes; AERDB_L2 combines the land DB algorithm with the SOAR ocean aerosol retrieval. VIIRS AOD at 6 km can resolve city-scale features and, combined with daily coverage, is suitable for regional air-quality monitoring and short-term pollution event tracking [38], [39], [40]. The long-term SNPP/NOAA20 series, together with MODIS, builds ~25 years of global aerosol records for climate studies.

3) MERRA-2 Dataset: The modern-era retrospective analysis for research and applications, version 2 (MERRA-2), developed by NASA's GMAO, is a global atmospheric reanalysis dataset. Its AOD product integrates multisource observations with an advanced atmospheric chemistry-transport model to provide spatiotemporally continuous aerosol distributions. In this study, daily mean MERRA-2 AOD for January 2024 at 550 nm is used as one of the auxiliary variables for modeling and to assist AOD imputation by leveraging its continuity; the spatial resolution is $0.5^\circ \times 0.625^\circ$. In addition, MERRA-2 AOD is cross-compared with the imputed AOD results to assess the reliability of satellite retrievals over complex surfaces—providing an independent reference especially over oceans and deserts [41], [42], [43].

It is noteworthy that within the framework of this study, MERRA-2 AOD does not participate directly in the OI fusion. Instead, it serves as one of the auxiliary variables in the transformer model, constituting a multivariate input together with ERA5 meteorological variables. As a reanalysis dataset, MERRA-2 offers the advantages of spatiotemporal continuity and no missing values. However, its relatively coarse spatial resolution and the fact that it has already partially assimilated satellite observations mean that its direct inclusion in the OI process could lead to information redundancy and excessive smoothing. Therefore, in this work, MERRA-2 AOD is positioned as a physically driven auxiliary feature rather than a prior field in the statistical fusion. This approach preserves the independence of the OI step while enhancing the model's capability for physically consistent reasoning over cloud-obscured and high-reflectance regions.

4) ERA5 Dataset: Fifth generation of European reanalysis (ERA5), produced by ECMWF, is a global atmospheric reanalysis that fuses massive observations with advanced numerical modeling to provide high-resolution, multivariable meteorological parameters. In this study, January 2024 ERA5 data are used to aid the construction of the AOD imputation

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model. We extract nine key variables: relative humidity at 1000/850/500 hPa, 10-m wind U and V components, 2-m air temperature, surface pressure, boundary layer height, and total precipitation; the spatial resolution is $0.25^\circ \times 0.25^\circ$. These variables jointly represent thermodynamic, dynamic, moisture, and boundary-layer processes, enabling a systematic assessment of meteorological influences on aerosol distributions (e.g., hygroscopic growth driven by humidity and advection driven by wind). ERA5's multielement, high-resolution, and complete spatiotemporal coverage complements satellite AOD and MERRA-2 AOD, improving the overall robustness of this study [44], [45], [46].

5) Aeronet Dataset: The AERONET, jointly established by the National Aeronautics and Space Administration (NASA) and the French National Centre for Scientific Research (CNRS), is a global ground-based aerosol monitoring network. It provides high-precision AOD measurements using standardized sun photometers, primarily the CE-318 model. Comprising nearly a thousand sites distributed worldwide, the AERONET network captures the spectral dependence characteristics of aerosols. The sites deliver accurate AOD measurements at approximately 15-min intervals across the 340–1640-nm range, with an uncertainty of about 0.01–0.02. These data are widely recognized as the "ground truth" for validating satellite-derived AOD estimates [47], [48], [49]. This study adopts Level 2.0 quality-assured AERONET data for January 2024 globally to validate satellite and reanalysis AOD. To match satellite bands, 550-nm AOD is obtained from AERONET's original channels using the Ångström-exponent–based band conversion; the band relationship follows:

$$\tau_a(\lambda) = k \cdot \lambda^{-a}. \tag{1}$$

In (1), k represents the atmospheric turbidity coefficient, which can be approximated by the AOD at 1020 nm. The Ångström exponent a is derived through nonlinear fitting of AOD values at five wavelengths (1020, 870, 670, 500, and 440 nm), yielding both k and a, which are then used to calculate AOD at 550 nm. This method effectively eliminates differences across spectral channels, enabling direct comparison of ground-based and satellite observations at a consistent wavelength.

Subsequently, temporal and spatial collocation between AERONET and satellite AOD products is performed. Spatially, each AERONET site is matched to its corresponding satellite grid cells by extracting a 3×3 neighborhood centered on the site, and the mean value of valid pixels within this window is compared with the ground measurement. Temporally, AERONET measurements are averaged over the local time window from 13:00 to 14:00, corresponding to satellite overpass times, to ensure consistency.

The global distribution of AERONET sites, covering both land and ocean, together with its unified observation standards and strict QC protocols, ensures consistency across sites worldwide. This provides a robust and comparable basis for validating satellite-derived AOD, thereby avoiding regional observational discrepancies and enhancing the global applicability of the validation results.

III. Methodology

This study focuses on the fusion and gap-filling of eight global AOD products derived from the MODIS and VIIRS sensors, with the aim of developing a globally applicable AOD imputation framework and producing seamless global AOD datasets. As illustrated in Fig. 2, the overall workflow consists of three stages. First, the eight satellite AOD products are preliminarily merged using an OI approach, which enhances both the spatial coverage and overall accuracy of the fused dataset while simultaneously reducing redundancy and data volume, thereby improving efficiency for subsequent large-scale imputation. Second, auxiliary datasets from MERRA-2 and ERA5 are preprocessed to ensure consistency in spatial and temporal resolution, and nine ERA5 meteorological variables are further subjected to feature extraction and dimensionality reduction. Finally, a transformer-based global seamless AOD gap-filling model is constructed, which leverages both the fused AOD products and auxiliary meteorological variables to train the model, leading to the development of a robust imputation framework capable of generating seamless global AOD fields. The detailed methodology is described below.

A. Data Preprocessing

To ensure spatiotemporal consistency and comparability among multisource datasets, a comprehensive preprocessing procedure was applied to both the reanalysis datasets (MERRA-2 and ERA5) and the eight MODIS/VIIRS AOD products. Reanalysis data are characterized by complete temporal coverage and spatial continuity, but at relatively coarse resolution; satellite products, in contrast, provide finer spatial detail but suffer from large areas of missing data due to cloud contamination and unfavorable surface conditions. Thus, effective integration of these two data types forms the foundation for generating seamless AOD products at the global scale.

To enable effective correlation analysis between satellite AOD products and reanalysis meteorological data, it is essential to first address the spatiotemporal heterogeneity among data from different sources. This is achieved through spatiotemporal registration, which aligns satellite products and reanalysis data to a unified temporal scale and spatial reference, thereby eliminating systematic biases for subsequent fusion and quantitative analysis. The core of temporal alignment lies in matching the temporal resolution of meteorological data with the observation frequency of satellite AOD, a design based on the orbital characteristics of polar-orbiting satellites and the inherent differences in their temporal resolutions. The three satellites used in this study (Aqua, SNPP, and NOAA20) all follow an "afternoon satellite" orbit design. Unlike geostationary satellites, which provide continuous all-weather observations, polar-orbiting satellites have periodic orbits that result in approximately one overpass per day for any given location on Earth. The three selected afternoon satellite products exhibit highly consistent overpass times for the same latitude-longitude grid points, with the nadir point concentrated around 13:30 local time. Since polar-orbiting satellites cover most global regions once daily, a daily scale was adopted as the unified temporal baseline for

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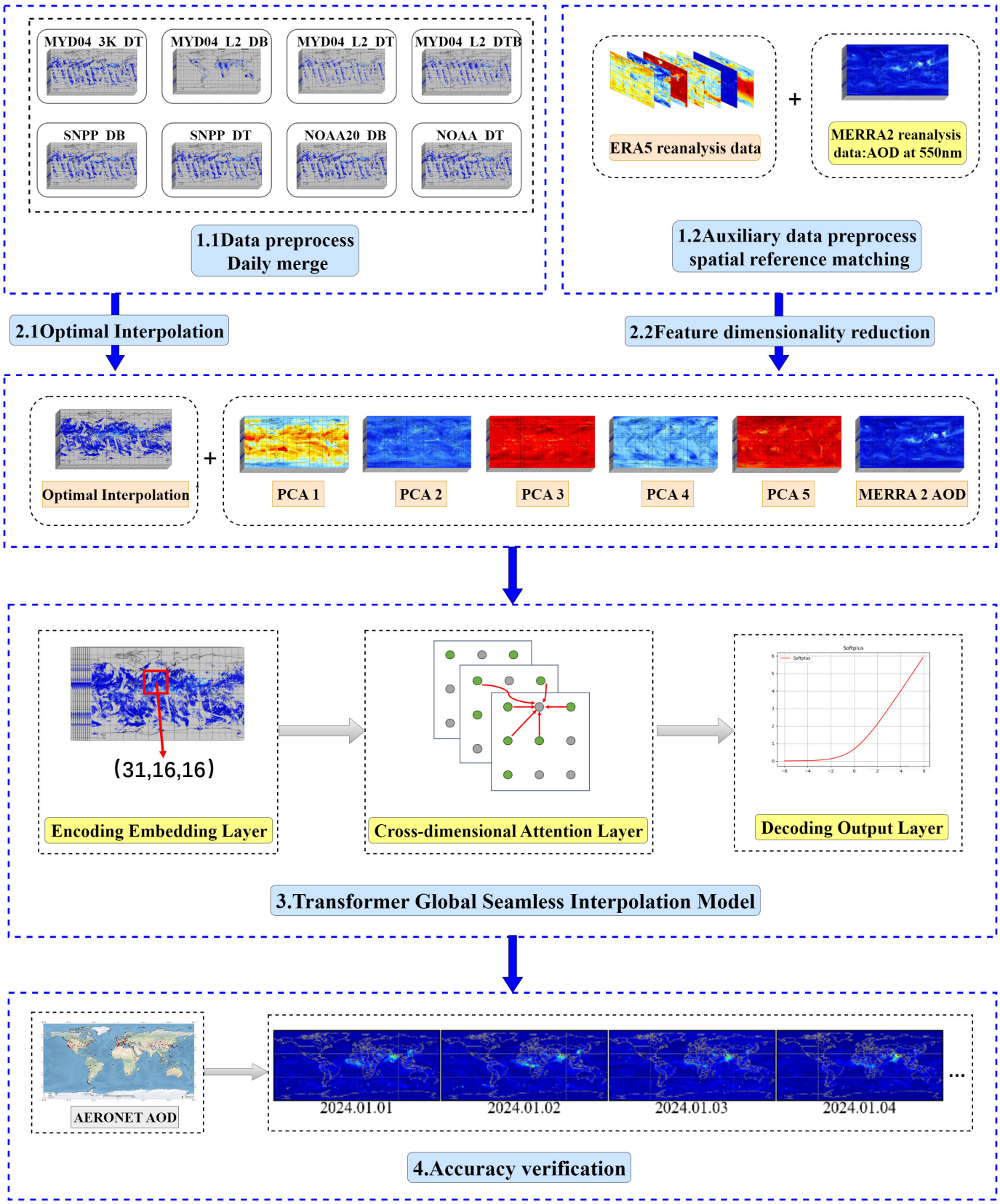


Fig. 2. Overall experimental workflow.

all data products. The temporal resolution of the reanalysis meteorological data is hourly; therefore, this study employed a temporal window averaging method centered on the satellite overpass time for temporal downsampling. Specifically, a 3-h

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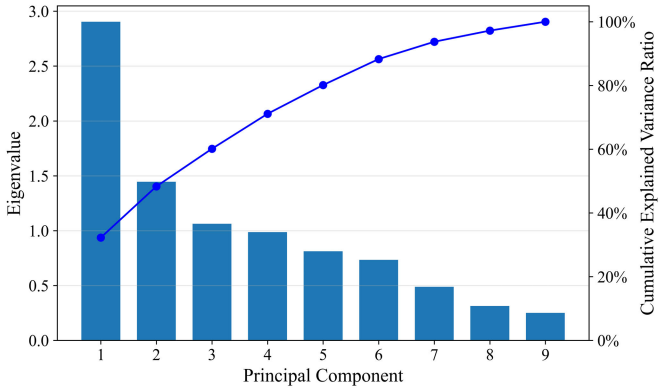


Fig. 3. Scree plot of the PCAs. The bar chart represents the eigenvalues of each principal component (values corresponding to the left vertical axis), and the line graph shows the cumulative variance explained by the principal components (values corresponding to the right vertical axis).

window (±1.5 h around the overpass time) was selected to encompass the near-real-time meteorological context during satellite observation. Arithmetic averaging was applied to the reanalysis meteorological data within this window to aggregate the data to a daily scale.

The objective of spatial registration is to eliminate resolution discrepancies among different datasets and achieve one-to-one correspondence between grid cells. The key lies in constructing a reference grid and selecting an appropriate spatial resampling method. Considering the spatial resolutions of the eight satellite AOD products, a global reference grid at $0.1^\circ \times 0.1^\circ$ was established. At the equator, this grid corresponds to an actual spatial distance of approximately 11×11 km. Toward mid and high latitudes, the actual east-west distance slightly decreases due to the convergence of meridians, but the overall scale remains compatible with the spatial resolutions of most satellite AOD products. All raster data underwent spatial registration through two steps: projection unification and resampling. Using the $0.1^\circ \times 0.1^\circ$ reference grid as the target, bilinear interpolation was applied for spatial resampling of each dataset. This method works by assigning weights to the four nearest original grid points around the target cell. The weights are based on distance. The target value is then computed through a weighted average. This approach effectively achieves spatial alignment among data from diverse sources and resolutions.

Furthermore, to reduce multicollinearity among auxiliary variables and extract dominant climatic signals, the nine typical meteorological variables from ERA5 were standardized and subjected to principal component analysis (PCA). As shown in Fig. 3, the first five principal components were selected based on the analysis results, collectively explaining over 80% of the cumulative variance. These components effectively represent major atmospheric circulation patterns and regional meteorological features and were used as auxiliary inputs in subsequent modeling. The loadings of each meteorological variable on the first five principal components are presented in Table II, illustrating the correlation degree between each variable and each principal component. In addition, all resampled satellite AOD products underwent bias

TABLE II
LOADINGS OF EACH METEOROLOGICAL VARIABLE ON THE FIRST FIVE PRINCIPAL COMPONENTS

	PC1	PC2	PC3	PC4	PC5
1000 hPa relative humidity	-0.722	0.258	-0.110	-0.024	0.512
850 hPa relative humidity	-0.570	0.508	-0.097	0.072	-0.055
500 hPa relative humidity	-0.737	0.266	0.064	0.019	-0.086
2 m temperature	0.822	0.283	-0.211	0.082	-0.030
10 m u-component of wind	-0.095	0.325	0.896	-0.068	-0.025
10 m v-component of wind	-0.058	-0.219	0.135	0.954	0.138
Boundary layer height	0.691	0.390	0.257	0.088	-0.125
Surface pressure	0.591	0.356	-0.014	-0.059	0.646
Total precipitation	-0.030	0.734	-0.323	0.218	-0.293

correction using AERONET site data. Specifically, correction coefficients a and b were derived via linear regression, and adjustments were made according to

$$AOD_{\text{calibrated}} = a \times AOD_{\text{satellite}} + b. \quad (2)$$

The purpose of this step is to minimize the impact of systematic biases inherent in the original satellite AOD products on subsequent fusion and imputation. To avoid compromising the use of AERONET Level 2.0 data for accuracy validation in later stages, the comprehensive bias correction herein utilized contemporaneous AERONET Level 1.5 data. Through the above standardized preprocessing pipeline, this study established a consistent multisource daily dataset at 0.1° resolution, laying a solid data foundation for subsequent OI fusion and transformer-based imputation.

B. OI Fusion

Although satellite-based AOD products provide relatively high spatial resolution, they suffer from extensive missing values due to cloud contamination and unfavorable sensor viewing conditions. Therefore, it is essential to fuse the eight selected AOD products in this study to generate a dataset with higher coverage and improved accuracy by leveraging the complementary information among multisource products.

The OI method is a widely used data assimilation technique in numerical weather prediction and environmental sciences. Its core principle is to estimate the optimal weight matrix using the error covariance of both the background field and the observations, thereby minimizing the error variance of the

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analysis field. Unlike simple weighted averaging or empirical regression, OI explicitly incorporates both observational and background errors, yielding an unbiased optimal estimate of the analysis field in a rigorous statistical sense.

For any grid cell to be estimated, the basic form of OI can be expressed as in the following equation:

$$x_a = x_b + K(y - Hx_b) \tag{3}$$

where x_a represents the fused value, x_b denotes the background field, y is the observation vector (i.e., the satellite AOD product), H is the observation operator, and K is the Kalman gain matrix, defined as in the following equation [50], [51], [52]:

$$K = BH^T \left(HBH^T + R\right)^{-1}. \tag{4}$$

Here, B and R are the background and observation error covariance matrices, respectively. Typically, R is diagonal, with diagonal elements representing the error variances of different data sources. Since all products have been co-registered during preprocessing, spatial displacement errors are negligible, and the background and observation fields are assumed to correspond to the same location. Under this assumption, the equations for two-product fusion can be simplified as

$$A = F + W(O - F) \tag{5}$$

$$W = B\left(B + R^{-1}\right)^{-1} R^{-1} \tag{6}$$

where B is the error variance of the background field, R is the error variance of the observation, O is the observed value, F is the background value, and A is the fused value. In the pixel-wise OI process implemented in this study, the selection of the background field and observation field strictly follows the principle of statistical optimality. Specifically, the number of valid observations N from the eight satellite AOD products is first counted for the target pixel. If $N \geq 2$, the product with the smallest error variance relative to AERONET observations is selected as the background field F for that pixel, serving as the most reliable prior estimate. The remaining valid observations are then sequentially integrated into the current analysis field in ascending order of their respective error variances. Each integration step treats the current optimal estimate as the new background field and the next product to be fused as the observation, updating the analysis field via (5) and (6) until all valid observations are incorporated. If $N = 1$, the single valid observation is directly assigned as the fusion result. If $N = 0$, the pixel remains marked as missing and is reserved for subsequent model-based imputation.

The error variance for each product is estimated entirely based on AERONET ground-based observations. By spatiotemporally matching the global Level 2.0 AERONET site measurements with the average value within a 3×3 pixel window centered on each site during the satellite overpass time window (13:00–14:00 local time), the error for each matched pair is computed, and the mean squared error is used to represent the global relative accuracy of each product. In determining the fusion weights, a composite weighting strategy as shown in (7) is adopted, incorporating both a precision term based on error variance and a spatial consistency term based on 3×3 neighborhood agreement. In the formula, σ_n denotes the standard deviation of valid AOD values within a 3×3 window centered at the target pixel. The two terms are normalized and multiplied to obtain the final weight. This design is grounded in the reasonable assumption that the error characteristics of each product remain relatively stable across the globe, thereby achieving statistically minimized error variance in the fusion results while fully leveraging the complementarity of multisource data. The process ultimately provides an initial AOD field with enhanced coverage and accuracy for the subsequent transformer model [53], [54], [55], [56]

$$\text{weight} = \left(\frac{1/\sigma_e^2}{\sum_{i=1}^{n}\left(1/\sigma_{e,i}^2\right)}\right) \times \left(\frac{1/\sigma_n^2}{\sum_{j=1}^{m}\left(1/\sigma_{n,j}^2\right)}\right). \tag{7}$$

The composite weighting scheme proposed in (7) builds upon the traditional OI—which relies solely on error variance—by introducing an additional spatial neighborhood consistency constraint, defined as the variance within a 3×3 window. This design offers a dual advantage: it preserves the minimum-variance estimation property of OI while penalizing observations with high local spatial heterogeneity, thereby reducing "speckle noise" caused by single-pixel outliers or co-registration errors. We validated the effectiveness of this approach through comparative experiments. Under identical inputs, the fusion results from traditional OI exhibited noticeable speckle noise, particularly in high-latitude and marginal zones, leading to spatial discontinuities. In contrast, the composite weighting scheme maintained global accuracy while significantly improving spatial smoothness and visual continuity. Such joint spatial-accuracy weighting strategies have been demonstrated in recent multisource remote sensing data fusion studies to outperform purely statistical weighting [53], [54], [55].

C. Construction of the Transformer-Based Imputation Model

Despite the improvements brought by OI, gaps still remain in the fused dataset, especially in regions frequently covered by clouds or with high surface reflectance. In cases where all satellite products fail to provide valid retrievals, the grid cells remain empty.

Ma et al. [57] introduced the cross-dimensional self-attention (CDSA) mechanism for spatiotemporal imputation and forecasting of multivariate georeferenced time series. The classic transformer employs a full self-attention mechanism, treating all input tokens as a 1-D sequence and computing attention weights between any two tokens. Although this design has shown outstanding performance in NLP and image classification, it does not explicitly account for the intrinsic heterogeneity of multidimensional data structures. When applied directly to spatiotemporal geospatial data with regular grid structures, it tends to overlook the distinct physical meanings across different dimensions. In contrast, the CDSA architecture explicitly respects the multidimensional nature of spatiotemporal data by decomposing attention computation into dimensions such as time and space [57]. Independent attention modeling is performed within each dimension, and cross-dimensional information is then fused through weighting

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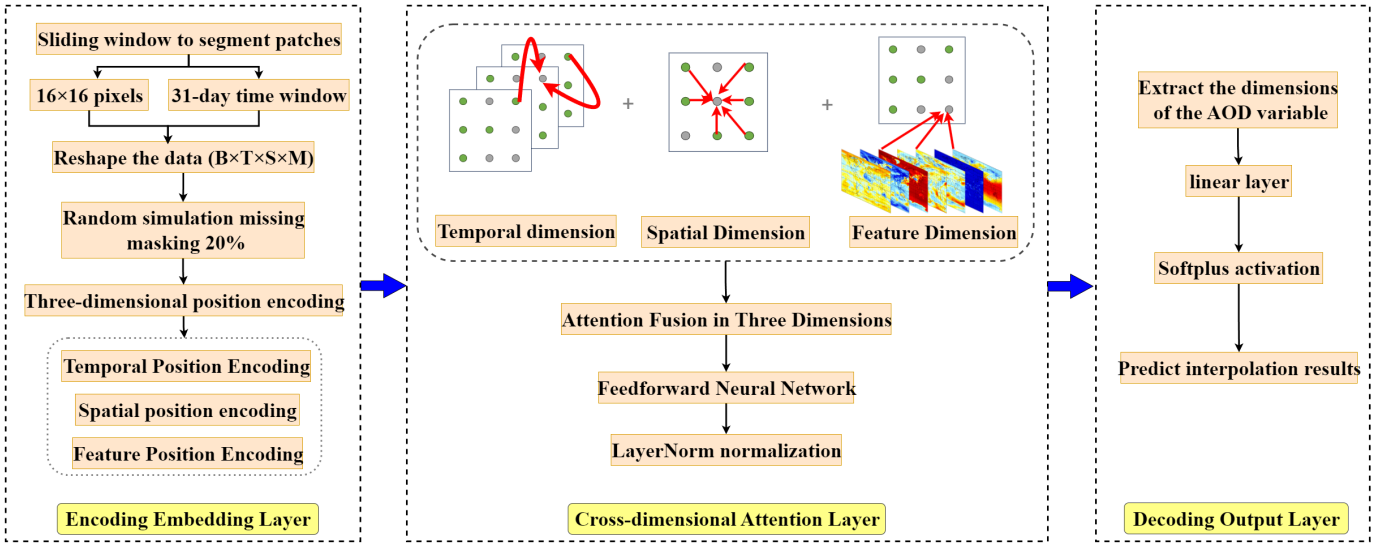


Fig. 4. Structure of the transformer-based imputation model.

or averaging strategies. This approach more naturally captures dependencies within each dimension and their interactions.

In terms of computational complexity, the classic transformer's quadratic complexity results in high memory and computational overhead when processing global high-resolution AOD grid data, limiting its application in large-scale spatiotemporal imputation tasks. By decomposing the attention computation, CDSA reduces the overall complexity to the sum of quadratic terms of each dimension. In the context of this study, CDSA significantly lowers computational burden while retaining the ability to model long-range dependencies.

Regarding applicability, the CDSA framework is specifically designed for geographically referenced multivariate spatiotemporal sequences, making it particularly suitable for reconstructing field data with regular grid structures in fields such as meteorology and remote sensing. In the task of AOD imputation under clouds in this study, CDSA can capture the diurnal evolution of aerosols along the time axis, enable cross-regional analog learning along the spatial axis, and integrate auxiliary information such as meteorological conditions along the variable axis, thereby achieving efficient and physically consistent imputation of cloud-covered areas.

Unlike traditional RNN-based methods, CDSA does not rely on sequential modeling. Instead, it explicitly captures long-range dependencies while improving training efficiency. CDSA computes attention maps separately for each dimension. It then combines these maps via tensor products. This approach preserves cross-dimensional dependencies but at a much lower computational cost than direct joint modeling. Experimental results have demonstrated that CDSA outperforms state-of-the-art approaches across multiple benchmark datasets, achieving higher accuracy in both missing-value imputation and long-term forecasting tasks.

Given that global AOD gridded products naturally conform to the characteristics of georeferenced spatiotemporal sequences, this study integrates the CDSA mechanism into a transformer-based AOD imputation framework, tailored to the characteristics of the data. The transformer model based on CDSA, as constructed in this study, operates on the core idea of leveraging valid spatiotemporal observation sequences around cloud-covered regions along with meteorological auxiliary variables to build a generative framework capable of modeling long-range dependencies, thereby achieving physically reasonable imputation of AOD under clouds. Specifically, the principle of filling cloud-obscured AOD values relies on spatiotemporal continuity modeling. Cloud systems generally exhibit spatiotemporal continuity, and AOD observations from clear periods before and after cloud coverage can provide important clues about the aerosol state beneath clouds. The self-attention mechanism in transformer can capture long-term dependencies, identify the evolution of AOD before and after cloud occurrence, and thus infer the aerosol load during cloudy periods. In addition to the AOD sequence, the model also incorporates nine types of meteorological variables provided by ERA5 and MERRA-2 AOD products, which are closely related to cloud formation, dissipation, and aerosol processes such as wet removal and transport. Through cross-dimensional attention, the model learns the coupling relationship between AOD and meteorological conditions, using meteorological fields as proxy variables for imputation in cloud-covered regions.

Specifically, to achieve seamless global AOD reconstruction, we extend the OI-fused dataset with a transformer model incorporating CDSA. The model is designed along three orthogonal axes: temporal (time), spatial (space), and variable (auxiliary features). Self-attention is computed separately along each dimension, followed by lightweight feature fusion and residual updates. This design approximates the modeling capacity of full spatiotemporal-variable joint attention with near-linear complexity, thereby significantly reducing memory and computational costs while retaining strong capability in capturing long-range dependencies and multisource interactions [58], [59], [60]. The overall structure of the proposed transformer-based imputation model is illustrated in Fig. 4.

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1) Encoding and Embedding Layer: The selection of key hyperparameters in this study aims to balance the model's representational capacity, computational efficiency, and the characteristics of global-scale data. The temporal window length (T = 31) covers a full calendar month (January 2024), enabling the model to simultaneously capture both diurnal aerosol variations (such as pollution episodes) and the monthly seasonal background. This setting aligns well with reanalysis data and provides sufficient temporal context for inferring AOD values under cloud cover. The spatial patch size (S = 16 × 16) is flattened into 256 spatial tokens, representing an optimal tradeoff between spatial contextual scope and computational complexity. At the equator, this corresponds to approximately 1.6° × 1.6° (~176 km^2), which is adequate to capture mesoscale aerosol spatial structures (e.g., urban plumes and regional transport) while keeping the per-patch attention complexity manageable. Employing a lower embedding dimension (d = 8) is central to the lightweight model design. This significantly reduces the total number of parameters while maintaining essential feature interaction capabilities, allowing the model to be trained on global data using a single 48-GB GPU. Experiments show that d = 8 achieves comparable validation root mean square error (RMSE) to d = 16 or d = 32, but with markedly improved training efficiency. We conducted grid search and ablation studies for these parameters. For instance, comparing T = 15 and 31 revealed that T = 31 offers slightly better validation RMSE and training stability. Testing patch sizes of 8 × 8, 16 × 16, and 32 × 32 demonstrated that 16 × 16 strikes the best overall balance between detail preservation and computational efficiency. Adjusting d across values of 4, 8, 16, and 32 led to the selection of d = 8 as an optimal compromise between efficiency and model depth.

Let the input tensor be $X \in \mathbb{R}^{B\times T\times S\times M}$, where B denotes the batch size, T is the temporal window length (set to 31 days in this study), S represents the number of spatial tokens (with each spatial patch being 16 × 16, flattened to S = 256), and M denotes the number of variables (consisting of one AOD channel and six auxiliary features). A linear embedding is then applied as

$$\mathbf{H}^{(0)} = \mathrm{Emb}\,(\mathbf{X}) \in \mathbb{R}^{B\times T\times S\times M\times d} \tag{8}$$

where d is the feature dimension (set to $d = 8$ for lightweight modeling). To enhance spatiotemporal and seasonal representation, three types of learnable positional encodings are incorporated as in the following equations:

$$\mathbf{H}^{(0)} \leftarrow \mathbf{H}^{(0)} + \mathbf{P}_t + \mathbf{P}_s + \mathbf{P}_m \tag{9}$$

$$\mathbf{P}_t \in \mathbb{R}^{1\times T\times 1\times 1\times d} \tag{10}$$

$$\mathbf{P}_s \in \mathbb{R}^{1\times 1\times S\times 1\times d} \tag{11}$$

$$\mathbf{P}_m \in \mathbb{R}^{1\times 1\times 1\times M\times d}. \tag{12}$$

This decomposed positional encoding injects location-specific information across temporal, spatial, and variable dimensions through broadcasting. For the lth layer input $H^{(l)}$, the model aims to reconstruct missing AOD values under the condition of retaining observed pixels unchanged. The output tensor is $\hat{Y} \in \mathbb{R}^{B\times T\times S}$, where the AOD channel (variable dimension) is extracted as the final prediction. A multihead attention mechanism serves as the core of the encoder, enabling the model to learn different dependency patterns: some heads may emphasize instantaneous correlations among variables, while others may capture long-range spatiotemporal relationships [61], [62], [63].

After flattening the 16 × 16 spatial patches into 256 spatial tokens, this study employs learnable spatial positional encoding to explicitly inject the positional information of each token within the original grid. Specifically, for each 16 × 16 patch, we map its pixels into a 1-D sequence and assign a unique, learnable d-dimensional vector (where d = 8 in this study) to each position in the sequence, as mathematically represented in (11). During the forward pass, this encoding is added to the input features via a broadcasting mechanism. Although the serialization process loses explicit 2-D coordinates, each spatial token retains the absolute positional information of its original grid location through its distinct positional encoding. During self-attention computation, the model learns the relationships between different positional encodings, thereby implicitly recovering spatial adjacency. For instance, the positional encodings of adjacent pixels gradually learn similar representations during training, enabling the model to assign higher attention weights to spatially neighboring tokens.

Furthermore, the CDSA mechanism adopted in this model further reinforces the preservation of spatial structure. Within the spatial attention submodule, the model explicitly computes pairwise attention weights among all 256 tokens along the spatial dimension, directly modeling the relationships between any two spatial locations—whether adjacent or not in the original grid. This design enables the model to capture not only local adjacency but also long-range spatial dependencies, such as similarities between cloud-free regions across cloud gaps.

2) CDSA Layer: Direct full attention over all ($T \times S \times M$) tokens incurs a computational and memory cost of $O((TSM)^2)$, which is prohibitive for global AOD imputation tasks. To address this, the CDSA strategy [57] is adopted, which computes self-attention separately along each dimension. This reduces the complexity to the sum of three quadratic terms while maintaining the ability to capture long-range dependencies across temporal, spatial, and variable axes. Specifically, attention outputs $\tilde{H}_t^{(l)}$, $\tilde{H}_s^{(l)}$, and $\tilde{H}_m^{(l)}$ are computed, reshaped back to $B \times T \times S \times M \times d$, and updated with normalization and residual connections. The three outputs are then fused via arithmetic averaging, avoiding the instability of multiplicative fusion for out-of-distribution samples, as shown in the following equation:

$$\mathbf{A}^{(l)} = \frac{1}{3}\left(\mathbf{U}_t^{(l)} + \mathbf{U}_s^{(l)} + \mathbf{U}_m^{(l)}\right). \tag{13}$$

We perform dimension-wise attention computation. For the input tensor $\mathbf{H} \in \mathbb{R}^{B\times T\times S\times M\times d}$, we first reshape it along the target dimension and then compute the standard-scaled dot-product attention

$$\mathrm{Attention}\,(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{14}$$

$$\mathbf{Q} = \mathbf{W}_q\mathbf{X} \tag{15}$$

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$$\mathbf{K} = \mathbf{W}_k \mathbf{X} \tag{16}$$

$$\mathbf{V} = \mathbf{W}_v \mathbf{X} \tag{17}$$

where $\mathbf{W}_q, \mathbf{W}_k$, and $\mathbf{W}_v$ are learnable projection matrices. Specifically, for the three dimensions: Temporal attention: reshape the input to $(T, B \times S \times M, d)$ to capture dependencies across all time steps; Spatial attention: reshape the input to $(S, B \times T \times M, d)$ to model relationships among all spatial locations (the flattened 256 tokens); variable attention: reshape the input to $(M, B \times T \times S, d)$ to learn interactions between AOD and the meteorological auxiliary variables.

Subsequently, the three attention outputs $\mathbf{U}_t, \mathbf{U}_s$, and $\mathbf{U}_m$ (each restored to the original shape $B \times T \times S \times M \times d$) are fused via arithmetic averaging, as expressed in (13). This averaging strategy is simple and stable, effectively integrating contextual information from different dimensions while avoiding the training instability that may arise from more complex fusion schemes such as concatenation or weighted summation [57].

A feedforward neural network is subsequently applied for nonlinear transformation and residual updating. In terms of complexity, the CDSA model requires $O(B \cdot S \cdot M \cdot T^2), O(B \cdot T \cdot M \cdot S^2)$, and $O(B \cdot T \cdot S \cdot M^2)$, significantly lower than the $O(B \cdot (TSM)^2)$ of full attention. This reduction is particularly advantageous in our case, where M and local T are small, but S is large.

3) Decoding and Output Layer: In the decoder, multihead attention mechanisms interact with the encoder output to extract spatiotemporal-variable dependencies learned by the model. The AOD target channel (indexed as 0 in the variable dimension) is selected for prediction. To enforce the nonnegativity of AOD as a physical constraint, a Softplus activation function is applied, ensuring that the model does not produce unphysical negative values under low-concentration background conditions.

4) Self-Supervised Masking and Loss Function: To ensure the generalizability of the model and the objectivity of evaluation, this study adopts a "self-supervised training–independent ground-truth validation" paradigm for dataset construction. Both the training and validation processes are based on the globally available, temporally complete, valid observation pixels (i.e., nonmissing values) from the optimally interpolated fusion products. All known satellite observations collectively form the source data pool for self-supervised learning. In each training sample, 20% of the valid pixels are randomly masked, and their corresponding true values serve as the validation target; the remaining 80% of unmasked valid pixels constitute the model input. This procedure is performed dynamically in every training iteration, ensuring that the training set (unmasked portion) and the validation set (masked portion) share the same spatiotemporal origin while remaining informationally isolated.

The test set consists of all real missing regions across the globe. The model's imputation results for these missing areas are not involved in the training or validation loops; instead, they are spatiotemporally collocated and compared with completely independent, high-precision AERONET ground-based observations, which serve as the objective benchmark for final performance assessment.

To guard against any form of information leakage, the following three-level safeguards were implemented.

1) Separation between training and validation phases. During the self-supervised training loop, the 20% masked pixels used for validation and the 80% unmasked pixels used as input originate from the same random partition within each sample. The model can only observe the unmasked portion in each forward pass, and gradient updates are based solely on the reconstruction error of the masked portion. This ensures information isolation between the training and validation procedures at the sample level.

2) Absolute isolation of test data. Throughout the entire training and validation phases, the model never accesses the real, large-scale missing regions. The final imputation task is applied to all missing pixels only after training is fully completed. This prevents the model from "memorizing" or "peeking at" any distributional characteristics of the test set during training.

3) Independence of the evaluation benchmark. The final performance report is strictly based on AERONET ground-based measurements. This dataset is independent of all satellite products used for training and fusion in terms of data sources, spatiotemporal scales, and measurement principles, thereby eliminating potential evaluation bias that could arise from data homology or nesting.

During training, a random masking strategy is applied for self-supervised learning. The optimally interpolated AOD values are treated as "pseudo-truth" for supervision. With a masking ratio $\rho = 0.2$, a binary mask $M \in \{0,1\}^{T \times S}$ is generated to randomly hide a portion of observed AOD inputs, forcing the model to reconstruct them based on spatiotemporal-variable conditions. To avoid penalizing native missing values, an effective supervision mask is applied to filter out invalid cases [64], [65], [66].

The selection of a masking ratio of $\rho = 0.2$ is based on considerations of suitability and balance. In self-supervised learning research, $\rho = 0.1$–0.3 represents a commonly adopted design range. A lower masking ratio (e.g., $\rho < 0.1$) would render the reconstruction task too simple, preventing the model from adequately learning cross-dimensional dependencies; conversely, an excessively high masking ratio (e.g., $\rho > 0.4$) would lead to insufficient input information, increasing training instability and convergence difficulty. Setting $\rho = 0.2$ strikes a balance between task difficulty and information sufficiency, an approach that has been adopted in several spatiotemporal imputation and prediction studies [57], [64].

In the global AOD data examined in this study, missing values under clouds often appear as continuous patches rather than completely random distributions. A $\rho = 0.2$ can be regarded as a simplified representation of such continuous gaps, which simulates scenarios of locally incomplete information while avoiding excessive masking that would hinder the model's ability to capture spatial continuity. This setting enables the model to learn both how to utilize locally valid pixels and how to infer from long-range spatiotemporal associations during training. We conducted an ablation comparison on $\rho = \{0.1, 0.2, 0.3, 0.4\}$ in preliminary experiments to validate the impact of this parameter choice on the final imputation

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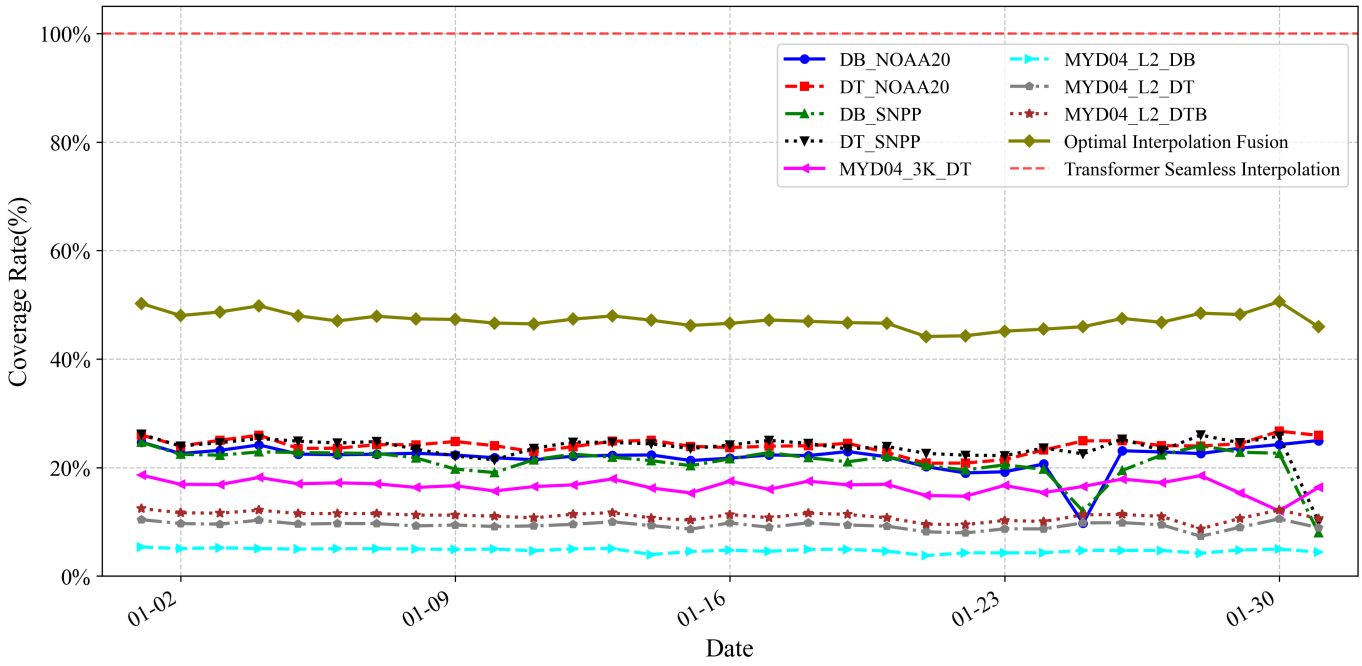


Fig. 5. Time series of coverage rates for the eight original AOD satellite products, the OI-fused product, and the transformer-imputed product in January 2024.

performance. The results indicate that $\rho = 0.2$ achieves the best balanced performance on the validation set, yielding optimal reconstruction error while maintaining higher training stability and faster convergence. Therefore, $\rho = 0.2$ was selected for this study.

5) Training and Inference: To ensure the reproducibility and transparency of the machine learning workflow, the key configurations for data construction, model training, and computational environment are detailed as follows. Samples were generated using a sliding temporal window of T = 31 days and 16×16 spatial patches, which were flattened into S = 256 tokens. The variable dimension comprised the target AOD channel and K selected reanalysis features, resulting in M = 1 + K input variables per spatiotemporal token. Computational efficiency was enhanced through delayed data loading, memory mapping, and the use of precomputed patch indices with cached masks.

The model was trained for 500 epochs with a batch size of 2. Optimization was performed using the AdamW optimizer with an initial learning rate of 5×10^{-4} and a weight decay coefficient of 0.01. Gradient clipping (max norm = 1.0) was applied to stabilize training, and no explicit learning rate scheduler was used. The loss function was defined as the RMSE, calculated exclusively over valid observed pixels. A self-supervised learning strategy was employed, where 20% of valid AOD observations were randomly masked during training to simulate missing data scenarios and encourage robust feature learning from spatiotemporal and meteorological contexts. To accelerate training and reduce GPU memory consumption, automatic mixed precision (AMP) training with gradient scaling was enabled. Model checkpoints were saved after each epoch, and the best-performing checkpoint was selected for final inference based on validation loss.

All experiments were conducted on a workstation equipped with an NVIDIA RTX A6000 GPU (48 GB of memory). Under this configuration, each training epoch required approximately 5 h, while a full global inference over the entire dataset was completed in about 0.3 h. These settings collectively ensure a stable and reproducible training process, supporting the reliability of the experimental outcomes.

D. Seamless AOD Imputation

At inference, the same input format is used as during training. The trained model performs block-wise predictions, and only missing pixels are replaced by model outputs. After sequential prediction and stitching, a seamless global AOD dataset is reconstructed.

1) E. Accuracy Assessment of Imputation:

To comprehensively evaluate the predictive performance of the proposed OI fusion and transformer-based imputation model for missing AOD values, this study employs global AERONET Level 2.0 observations as the ground truth for validation. Five statistical indicators are selected, namely, the correlation coefficient (r), coefficient of determination (R^2), RMSE, bias, and expected error (EE). The accuracy assessment procedure adopted in this study verifies the performance of different interpolation and fusion strategies through multimetric comparisons. At the same time, we not only focus on the overall global accuracy but also perform stratified evaluations under different aerosol types and surface conditions, in order to comprehensively reveal the applicability and limitations of the model.

In this experiment, the calculation equation for r is shown in (18), which reflects the consistency and linear correlation

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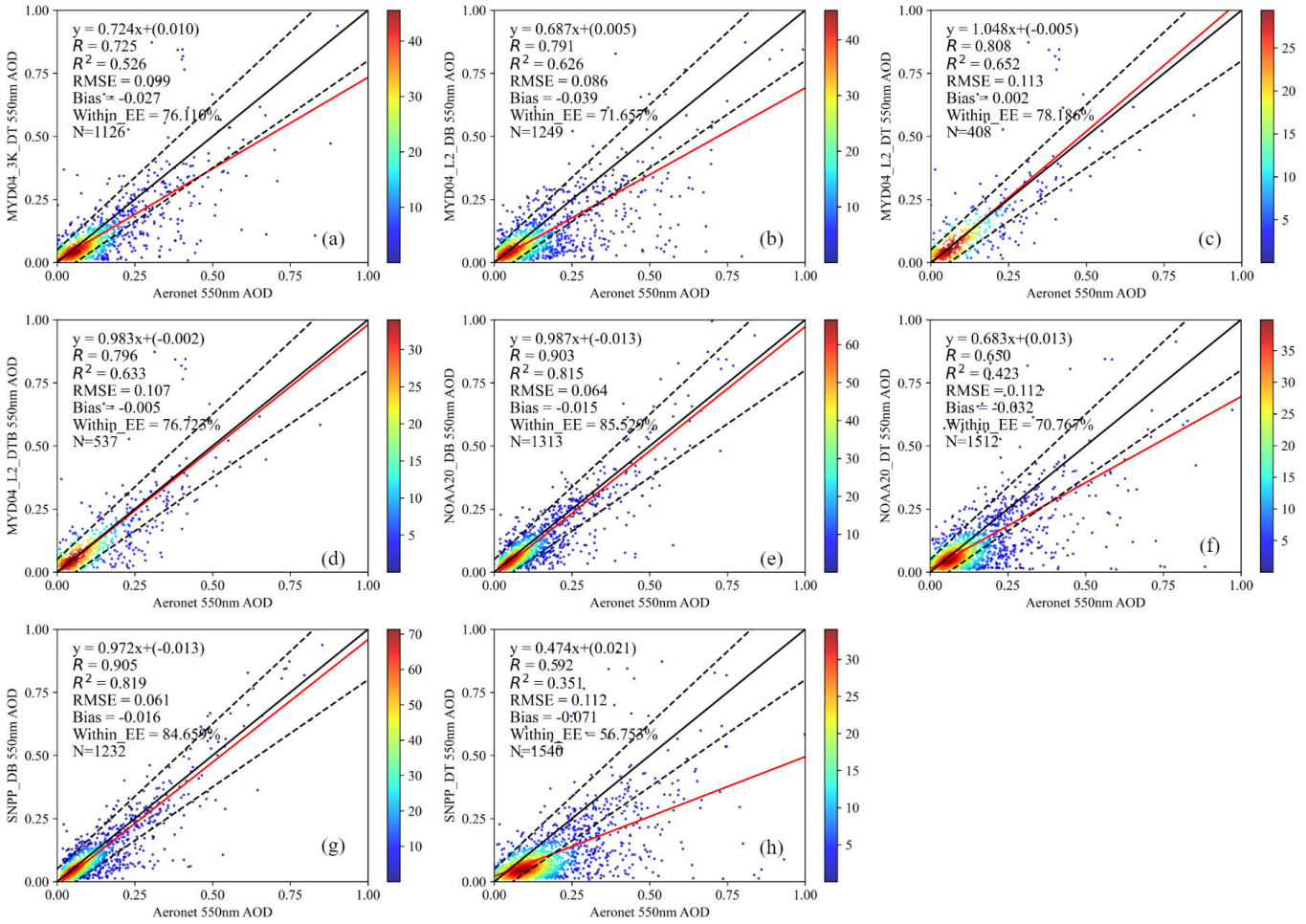


Fig. 6. Validation of satellite AOD products against AERONET observations during the study period. Subplots show scatterplots and regression fits for (a) MYD04_3K_DT, (b) MYD04_L2_DB, (c) MYD04_L2_DT, (d) MYD04_L2_DTB, (e) NOAA20_DB, (f) NOAA20_DT, (g) SNPP_DB, and (h) SNPP_DT. The red solid line indicates the regression fit, the black solid line is the 1:1 reference line, and the black dashed lines denote the EE envelope. The color scale represents sample density, with red indicating higher density and blue lower density.

between the model estimates and ground-based observations. The equation for R^2 is given in (19), which measures the ability of the model to explain the variance in the observations. The equation for RMSE is shown in (20), which indicates the average deviation of the model estimates from the observations. Bias is calculated using (21), which is used to determine whether the model systematically overestimates or underestimates AOD. For EE, we adopt the commonly used EE equation in AOD retrieval and evaluation, as shown in (22). Here, EE is defined as a dynamic error threshold that is typically used to determine whether the predicted results fall within an acceptable error range, and is one of the most widely used reference standards for satellite AOD products. During the statistical process, we further calculate the proportion of imputed results that fall within the EE range, which provides an intuitive measure of the reliability and practicality of the proposed imputation model [15], [23], [67]

$$R = \frac{\sum \left(AOD_{\text{model},i} - \overline{AOD}_{\text{model}}\right)\left(AOD_{\text{aeronet},i} - \overline{AOD}_{\text{aeronet}}\right)}{\sqrt{\sum \left(AOD_{\text{model},i} - \overline{AOD}_{\text{model}}\right)^2} \cdot \sqrt{\sum \left(AOD_{\text{aeronet},i} - \overline{AOD}_{\text{aeronet}}\right)^2}} \tag{18}$$

$$R^2 = 1 - \frac{\sum \left(AOD_{\text{aeronet},i} - AOD_{\text{model},i}\right)^2}{\sum \left(AOD_{\text{aeronet},i} - \overline{AOD}_{\text{aeronet}}\right)^2} \tag{19}$$

$$RMSE = \sqrt{\frac{1}{n}\sum \left(AOD_{\text{aeronet},i} - AOD_{\text{model},i}\right)^2} \tag{20}$$

$$\text{Bias} = \frac{1}{n}\sum \left(AOD_{\text{model},i} - AOD_{\text{aeronet},i}\right) \tag{21}$$

$$EE = \pm\left(0.05 + 0.15 \times AOD_{\text{aeronet}}\right) \tag{22}$$

In these equations, AOD_{model} represents the AOD value estimated by the model or satellite product, and AOD_{aeronet} represents the ground-based AERONET observation. During validation, the imputation results are first collocated with AERONET sites, and the average value of a 3×3 grid centered on each site is taken as the model estimate.

It is essential to objectively acknowledge that the spatial distribution of AERONET sites is notably uneven, as illustrated in Fig. 1. This spatial imbalance may introduce certain biases into the representativeness of global validation results. AERONET sites are predominantly clustered in regions with intensive research activities, such as North America, Europe, and East Asia, while being sparse in the Southern Hemisphere, over oceans, inland Africa, high-latitude areas, and the Tibetan Plateau. Such clustering leads to a highly concentrated spatial sample for validation, which may cause global statistical metrics (e.g., R^2 and RMSE) to predominantly reflect model performance in densely monitored regions, while inadequately

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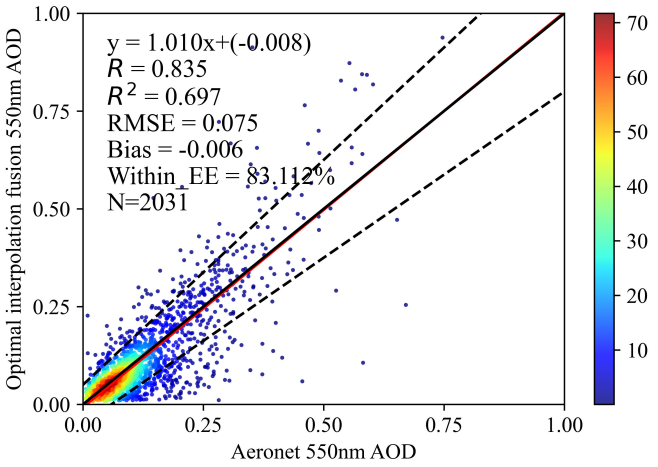


Fig. 7. Accuracy validation of the OI-fused product against AERONET AOD observations over the global domain during the experimental period. The red solid line indicates the regression line between the validated product and AERONET observations; the black solid line represents the 1:1 reference line; the area between the black dashed lines denotes the EE envelope; the color bar reflects the density of AOD matchups, with red indicating higher density and blue indicating lower density.

capturing error characteristics in sparsely covered areas. For example, the model's performance over oceans or deserts with few sites may not be effectively characterized. Furthermore, most sites are located in urban, suburban, or relatively open plain areas, offering limited representation of complex underlying surfaces—such as dense forests, snow-covered zones, and steep terrain—as well as regions with homogeneous aerosol types, such as clean marine backgrounds and dust source regions. This may result in evaluation blind spots over areas with high surface albedo and high aerosol model uncertainty, thereby affecting a comprehensive assessment of the model's robustness across diverse global environments.

To mitigate the potential impact of this uneven distribution, several measures were adopted in this study. First, all globally available Level 2.0 AERONET sites during the entire study period (January 2024) were incorporated, aiming to maximize spatial coverage. Second, stratified validation and regional analyses were conducted. Beyond global metrics, comparative assessments across different latitudinal zones and land–sea regions (see Section IV-D) were performed to reveal spatial heterogeneity in model performance. Third, a multimetric evaluation approach was applied. In addition to relying on R^2 and RMSE, metrics such as bias and the proportion within the dynamic EE envelope were jointly used to comprehensively reflect model performance across different AOD ranges.

Nevertheless, the uneven distribution of AERONET sites remains an inherent limitation in current global aerosol remote sensing validation. Future efforts could integrate other ground-based observational networks (e.g., SKYNET), shipborne or mobile observations, and high-resolution regional model simulations to construct a more balanced validation dataset, thereby further enhancing the robustness and representativeness of global-scale evaluations.

IV. Experimental Results and Discussion

A. Validation of Eight Satellite AOD Products

As illustrated in Fig. 5, we compare the daily coverage rates of the eight original satellite AOD products, the fused product from OI, and the seamless product generated by the transformer-based imputation model, where each line is plotted in a different color. The coverage rate in this study is defined as the fraction of valid AOD retrievals between 60°N and 60°S. As shown, the effective coverage of individual satellite products is generally low, typically ranging from 5% to 25%, with noticeable fluctuations in some cases. Each product exhibits strip-like gaps and regional missing values under varying cloud cover, surface reflectance, and orbital conditions. Overall, the four VIIRS-derived products show higher coverage than the four MODIS-derived products, though their stability is weaker. In late January, multiple VIIRS products suffered abrupt drops in daily coverage due to orbital issues. Therefore, effective fusion of the eight products is essential to improve both spatial coverage and temporal stability, thereby fully leveraging their complementary advantages.

In Fig. 6, we further validate the eight satellite AOD products against ground-based AERONET observations. Scatter plots with regression statistics are provided, along with calculated metrics, including R^2 and RMSE. Overall, the SNPP_DB and NOAA20_DB products, derived from the VIIRS sensor using the DB algorithm, achieve the best agreement with ground observations under large-sample conditions. Both exhibit correlation coefficients (R) above 0.9, determination coefficients (R^2) above 0.8, and RMSE values of approximately 0.06, representing the most accurate results among the eight products. In contrast, the corresponding DT products from the same platforms perform notably worse, with R values around 0.6 R^2 values around 0.4, and RMSE values reaching 0.112. These results reflect the inherent limitations of the DT algorithm compared with the DB algorithm.

Specifically, the DT algorithm, as an earlier retrieval approach, performs reasonably well over oceans and dense vegetation, but its applicability over bright or heterogeneous surfaces is limited, often resulting in degraded performance. By contrast, the DB algorithm incorporates the DB band and utilizes a surface reflectance database, thereby overcoming key bottlenecks in retrieval over bright and complex surfaces. Consequently, global-scale validation demonstrates that VIIRS DB products are significantly more accurate than DT products, highlighting the robustness of the DB algorithm under high-reflectance surface conditions.

For the MODIS sensor, the four products fall between the performance levels of VIIRS DB and DT products. Although not as outstanding as VIIRS DB, the MODIS products are generally more stable. For example, the MYD04_3K_DT and MYD04_L2_DT products, despite being more prone to systematic biases under cloud contamination and extreme viewing geometries, still achieve R values of $0.7 - 0.8$ and R^2 values of $0.5 - 0.6$, with the proportions of retrievals falling within the EE envelope reaching 76.110% and 78.186%, respectively. These results indicate higher accuracy than VIIRS DT products. The MYD04_L2_DTB product partially reduces

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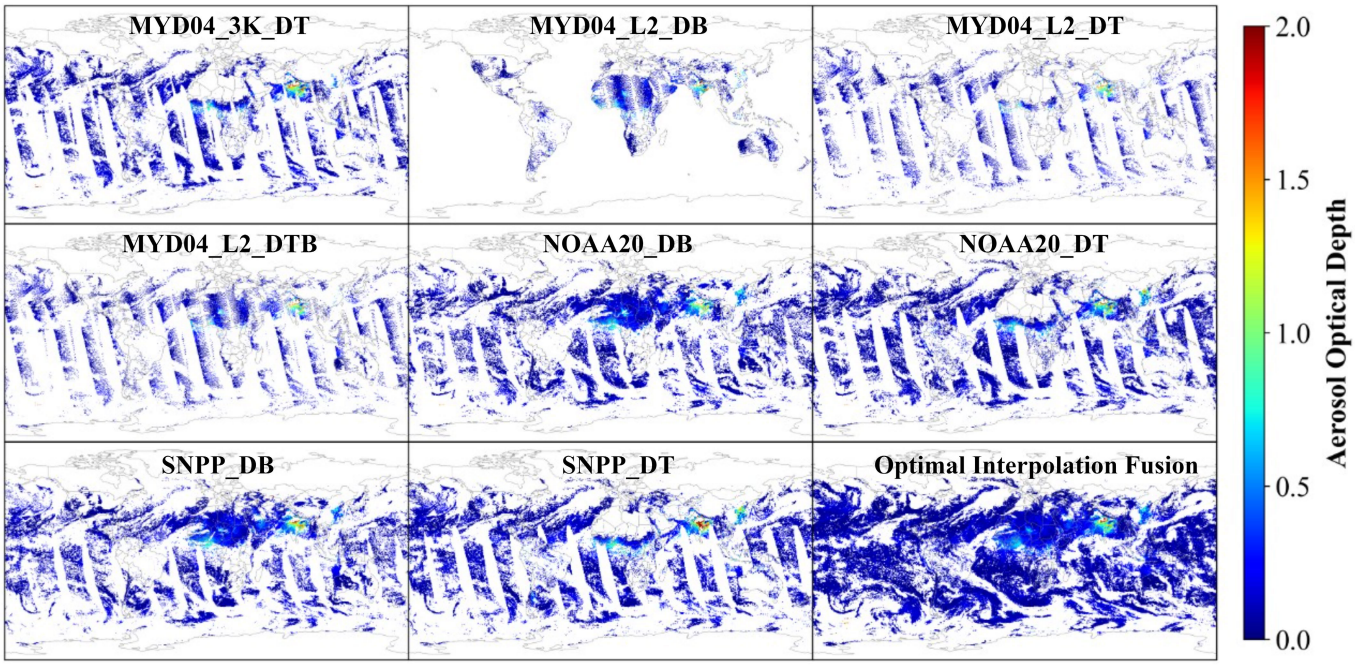


Fig. 8. Global AOD maps on January 1, 2024, for the eight original AOD satellite products and the OI-fused product. The color bar indicates AOD values, and white represents missing values.

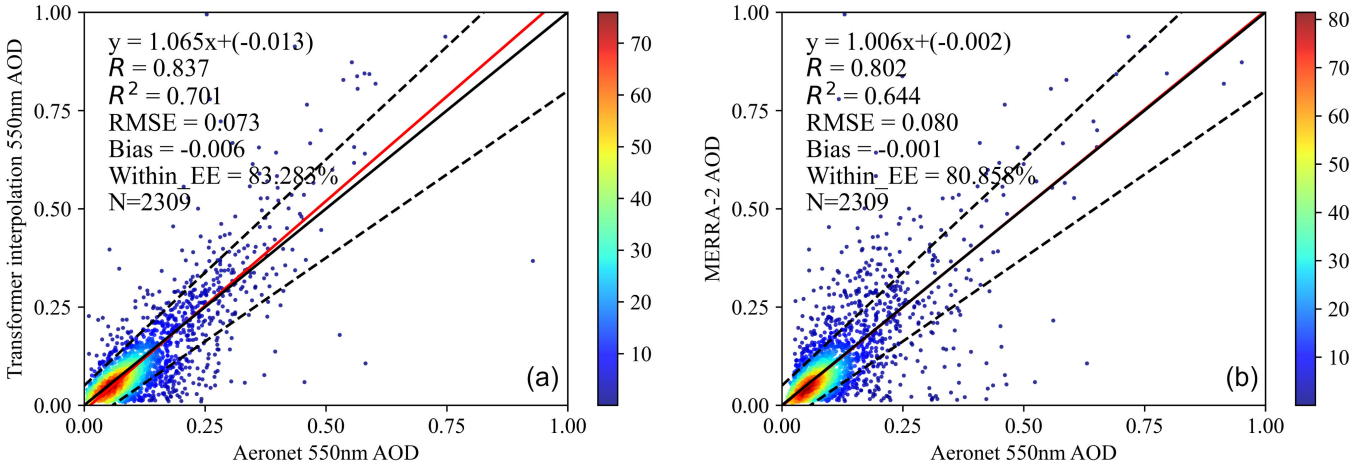


Fig. 9. Accuracy validation of (a) transformer-imputed seamless product and (b) MERRA-2 AOD product against AERONET AOD observations over the global domain during the experimental period. The red solid line represents the regression line between the validated product and AERONET; the black solid line is the 1:1 reference line; the area between the black dashed lines indicates the EE envelope; the color bar reflects the density of AOD matchups, with red indicating higher density and blue indicating lower density.

systematic errors but still shows clusters of outliers, whereas the MYD04_L2_DB product provides the most balanced performance among MODIS datasets. With R^2 exceeding 0.6, 1249 collocations with AERONET, and an RMSE of 0.086, MYD04_L2_DB is the most accurate MODIS product.

From the perspective of EE proportions, most products fall within the range of 57%–85%, with DB-series products consistently achieving higher ratios. This implies that during the local overpass times of Aqua, SNPP, and NOAA20 satellites (approximately 13:30 local time), a substantial proportion of pixels meet the acceptable error threshold for practical applications. Nevertheless, the spatial completeness of individual products remains insufficient due to cloud contamination and orbital geometry constraints, further underscoring the necessity of subsequent fusion and imputation. These findings also highlight the complementary nature of single-source AOD products and the importance of their integration.

B. Accuracy Assessment of OI Fusion Results

To validate the accuracy of the OI fusion, Fig. 7 presents the overall agreement between the fused product and AERONET observations. The OI-fused product achieves a more balanced accuracy and consistency across the full sample. First, through bias correction preprocessing applied to the original satellite products prior to fusion, systematic errors in the raw products were mitigated. The fused product exhibits a regression slope

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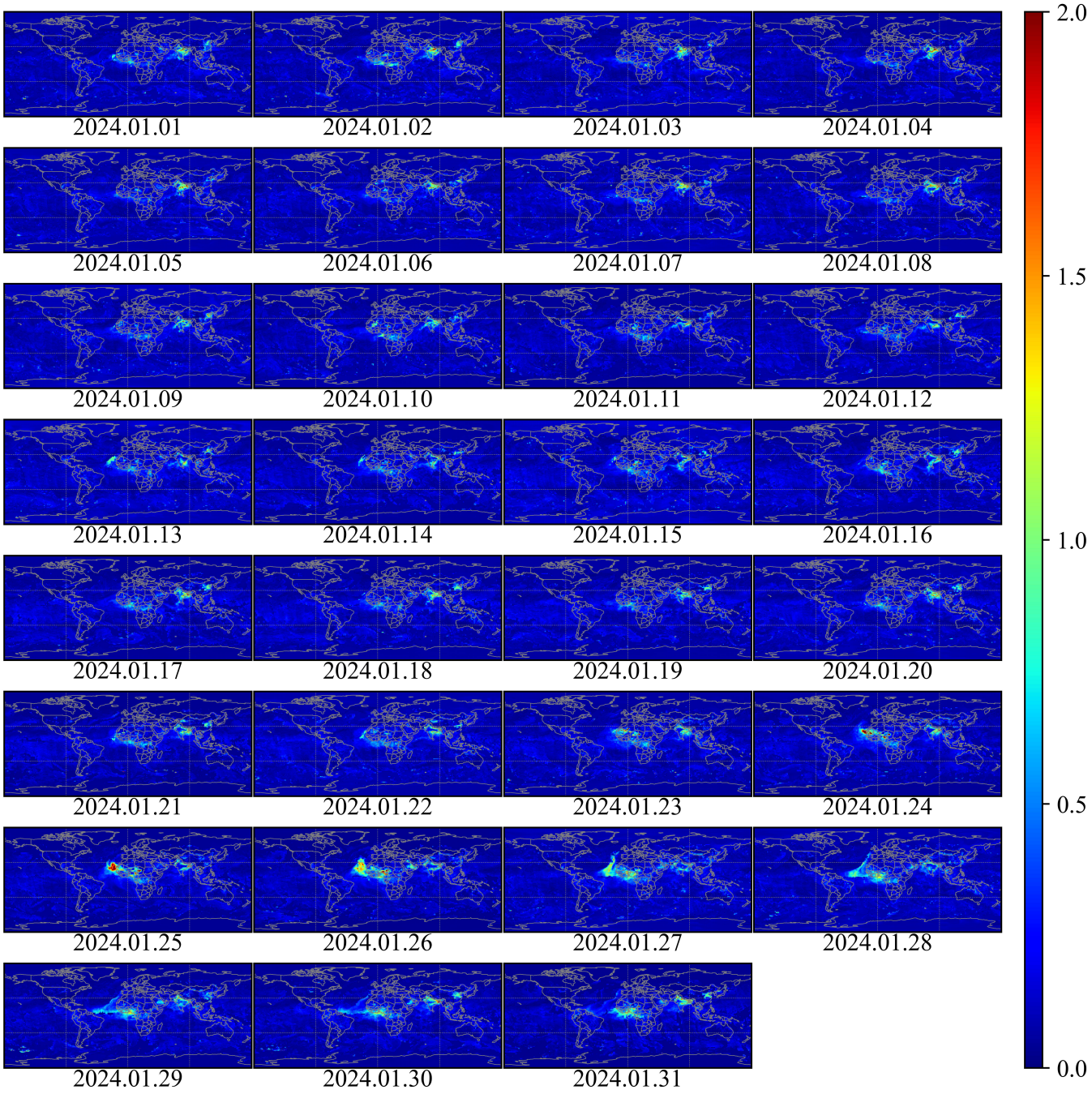


Fig. 10. Daily global AOD maps for January 2024. The color bar indicates AOD values.

close to 1 against AERONET, with a correlation coefficient R of 0.835 and R2 of 0.697. In terms of error and bias, the RMSE is 0.075, the bias is −0.006, and approximately 83% of the samples fall within the EE envelope, with a total sample size of 2031. Compared with the accuracy of the eight original products shown in Fig. 6, the fused product performs second only to the two best-performing products (SNPP_DB and NOAA20_DB) and surpasses the other six products, confirming the rationality of fusing eight different products. Moreover, the fusion results significantly reduce random error and systemic bias, with the scatter points clustering more closely around the 1:1 line compared to single-source products. This improvement demonstrates that by modeling and weighting the error characteristics of multisource products and incorporating spatial covariance constraints within geographic neighborhoods, the OI method achieves statistically superior AOD estimates.

The red curve in Fig. 5 illustrates the coverage of the OI-fused product. In contrast to the original satellite products, the fused product shows a significant improvement in coverage, maintaining stable daily coverage at approximately 50%. The enhancement in coverage achieved through fusion is more

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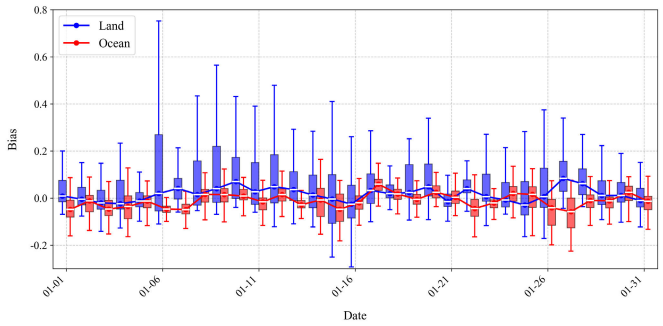


Fig. 11. Daily bias boxplots between the transformer-imputed product and the MERRA-2 product. Blue represents the land region; red represents the ocean region. The line within the box indicates the median; the top and bottom boundaries of the box represent the upper and lower quartiles, respectively. The whiskers indicate the data range. The vertical axis (bias) represents the difference between the transformer-imputed product and the MERRA-2 product.

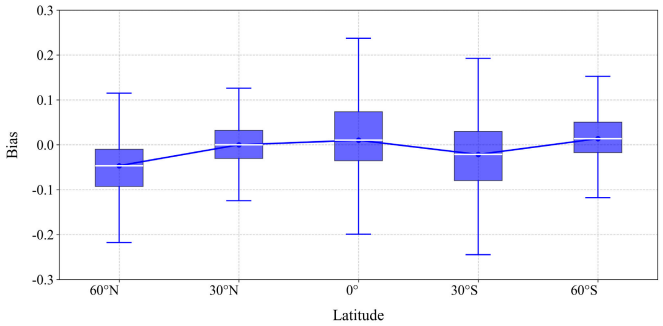


Fig. 12. Bias boxplots between the transformer-imputed product and the MERRA-2 product across different latitudes. The line within the box indicates the median; the top and bottom boundaries of the box represent the upper and lower quartiles, respectively. The whiskers indicate the data range. The vertical axis (bias) represents the difference between the transformer-imputed product and the MERRA-2 product.

intuitively demonstrated in Fig. 8. Using data from January 1, 2024, as an example, Fig. 8 displays the global coverage and AOD values of the eight original satellite products and the OI-fused product. Evidently, all original products exhibit notable strip-like and regional gaps, particularly over high-albedo surfaces such as deserts and snow-covered regions. In addition, the performance of different retrieval algorithms varies significantly across different land cover types. The fused result shows more continuous distributions and smoother transitions, especially along strip boundaries, while preserving the contrast between major pollution belts and regionally clean backgrounds. This indicates that the fusion achieves a reasonable balance between smoothing and preserving details.

The OI fusion effectively mitigates inconsistencies caused by strip gaps and algorithmic differences among single-source products, significantly enhancing coverage and stability without sacrificing accuracy. Overall, the large gaps in original satellite products are primarily attributable to three factors: cloud cover, satellite orbital configurations, and limitations of retrieval algorithms. The OI fusion integrates multiple products from different satellites and algorithms, thereby minimizing the impact of the latter two factors. After fusion, the remaining gaps—mainly due to cloud cover—are subsequently filled using the proposed transformer-based imputation model to achieve seamless reconstruction.

C. Accuracy Assessment of Transformer-Based Imputation Results

Using the transformer model for imputation, we obtain a globally seamless AOD product. To evaluate the accuracy of the product, we compare it with the MERRA-2 global reanalysis AOD product. As shown in Fig. 9, (a) and (b) present the overall accuracy of the transformer-imputed product and the MERRA-2 reanalysis product against AERONET, respectively. From Fig. 9(a), the statistical metrics show that R = 0.837 and R2 = 0.701, indicating slight improvements over the OI-fused results. Meanwhile, the RMSE and bias are 0.073 and –0.006, respectively—showing minor enhancements while remaining largely comparable to the OI-fused results. This demonstrates that imputation based on temporal, spatial, and auxiliary variables not only fills gaps but also improves the overall product accuracy.

In terms of sample size, the number of available samples increased from 2031 for the OI-fused product to 2309 for the seamless imputation product. Although seamless imputation significantly improves spatial coverage, the number of matchups with AERONET does not increase substantially.

This is understandable since the gaps filled during the transition from OI-fused to seamless global coverage are primarily caused by cloud cover and extreme weather conditions. Under such circumstances, AERONET sites are often unable to obtain valid high-quality measurements either. Therefore, imputing these regions does not significantly increase the number of AERONET matchups. To better validate the imputation accuracy of the transformer model, we further introduce the MERRA-2 reanalysis global AOD product for cross-validation.

In this section, we compare the transformer-imputed product with the MERRA-2 AOD product. As shown in Fig. 9, comparing (a) and (b), although the MERRA-2 AOD product benefits from reanalysis modeling and performs well in terms of bias and regression fit, the transformer-imputed product outperforms MERRA-2 in all four metrics: R, R2, RMSE, and the fraction within EE. In other words, the transformer-imputed product exhibits better correlation, variance explanation, and error performance than the MERRA-2 reanalysis product, with smaller deviations in medium-to-high AOD ranges. This indicates that by using the fused product as a prior and leveraging CDSA to jointly model temporal, spatial, and variable axes, the model more effectively utilizes reanalysis meteorological conditions and spatiotemporal neighborhood constraints, showing strong recovery capability for meteorological processes such as cloud dynamics.

As indicated by the red dashed line in Fig. 5, after seamless imputation via the transformer model, coverage reaches 100%, achieving spatiotemporally continuous daily full coverage. Compared to the original satellite products, global coverage increases from below 25% to 100%. Fig. 10 presents the daily seamless imputation results of global AOD for January 2024. First, the spatiotemporal continuity of the fully

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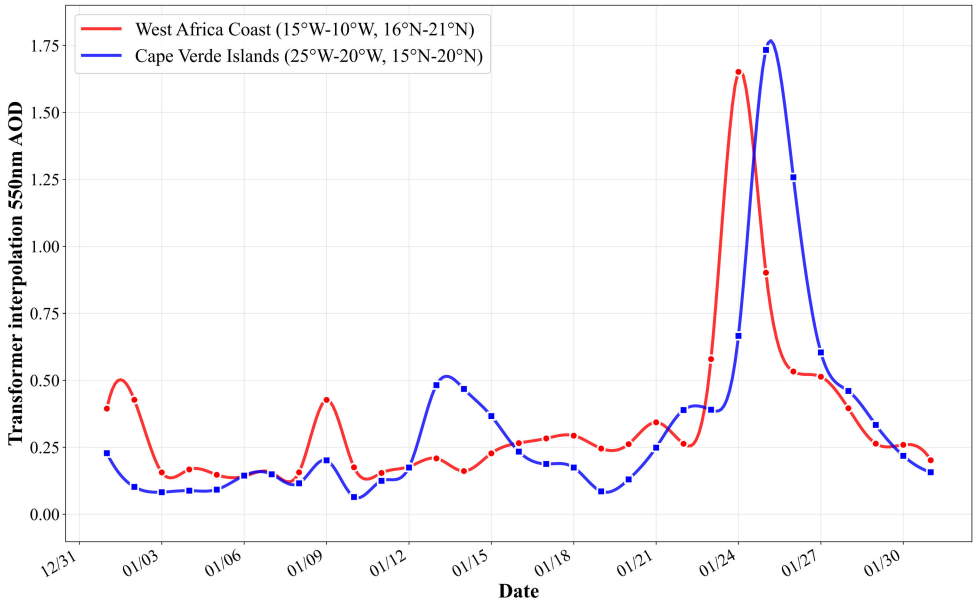


Fig. 13. Time series of AOD over key regions during a Sahara dust event in January 2024. The red line denotes the dust source region along the West African coast, and the blue line represents the Cape Verde Islands and the adjacent oceanic area.

imputed product is significantly enhanced. Regional pollution cores evolve smoothly across days without "interdaily jumps," such as the persistent pollution over India throughout the month and the Central African pollution belt in late January. Meanwhile, the anomalous increase and subsequent decline of AOD in the East Asian monsoon region in mid-to-late January are continuously captured, with the process from peak to decay consistent with synoptic conditions. The banded low-value zones in mid-latitude oceans of the Northern Hemisphere, influenced by frontal systems and westerlies, exhibit stable zonal structures.

Compared to previous interpolation models, the transformer-based imputation model achieves structured recovery in gap regions. By leveraging cross-dimensional attention to identify the most relevant conditional pixels across days and regions within a temporal window, it fully utilizes every valid observation. Moreover, constrained by reanalysis variables, the model captures advective transport and boundary layer evolution, not only completely filling gaps along land–sea boundaries and high-albedo regions but also imputing large missing areas such as high-altitude zones, deserts, and snow-covered regions based on trends in auxiliary variables. Through transformer-based imputation, a globally comprehensive spatiotemporal product with full coverage is obtained.

D. AOD Estimation Under Different Scenarios

We further conducted cross-validation between our seamless imputation product and the MERRA-2 AOD product to evaluate AOD estimation under different land cover types and latitudes. A land region (25°N–35°N, 100°E–110°E) and an ocean region (25°N–35°N, 150°E–160°E) within the same latitudinal band were selected to compare the impact of different underlying surfaces on AOD estimation. As shown in Fig. 11, the estimation bias between the transformer-imputed product and the MERRA-2 product is smaller over the ocean than over land. Over the ocean, the biases between the two products are mostly within ± 0.1, with the transformer-imputed product generally underestimating relative to MERRA-2. In contrast, over land, the discrepancies between the two products increase significantly, with a few bias values exceeding 0.2. Moreover, over land, the transformer product tends to overestimate compared to MERRA-2. Over land, the transformer product has access to more satellite measurements. This leads to more accurate AOD predictions. In comparison, the MERRA-2 product, being primarily based on an atmospheric chemical transport model, struggles to capture the complexity of real-world surface conditions. Over the ocean, however, the distribution and diffusion of atmospheric constituents are more stable due to minimal human activity, allowing the reanalysis model to capture the AOD state more accurately, and thus showing better agreement with our imputed product.

In addition, we selected data from five different latitudinal zones near 170°E to analyze the performance of the AOD imputation model under different thermal and climatic conditions. The selected latitudinal zones were 60°N, 30°N, the equator, 30°S, and 60°S. Since our experimental data cover January 2024, which corresponds to summer in the Southern Hemisphere and winter in the Northern Hemisphere, we were able to observe estimation trends under different seasonal conditions. As shown in Fig. 12, at 60°N, the transformer product shows a general underestimation compared to MERRA-2, while no systematic bias is observed in other latitudinal zones. The best agreement is achieved at 30°N, where the bias is concentrated within ± 0.03 and is overall unbiased. Slight and unstable biases are observed in the Southern Hemisphere and equatorial zones. It can be inferred that higher temperatures lead to more active atmospheric circulation, complicating AOD estimation. In contrast, lower temperatures generally

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stabilize AOD predictions, although extremely low temperatures may again introduce complexity.

To evaluate the capability of the proposed reconstruction framework in capturing representative pollution episodes, a significant transboundary Sahara dust event in January 2024 was analyzed (MODIS image gallery, accessed Jan 27, 2024 https://modis.gsfc.nasa.gov/gallery/individual.php?db_date=2024-01-27). Fig. 13 displays the AOD time series for two critical regions during this event: the dust source region along the West African coast (15°W–10°W, 16°N–21°N, red line) and the downwind area encompassing the Cape Verde Islands and nearby ocean (25°W–20°W, 15°N–20°N, blue line).

The temporal evolution reveals multiple pulsed AOD peaks in both regions throughout January, characteristic of the intermittent nature of dust outbreaks. As the primary source, the West African coast exhibits larger and more frequent AOD fluctuations. Pronounced peaks occurred around January 24 and 27, likely driven by meteorological conditions such as strong surface winds and dry convection, which trigger dust emission and uplift. In contrast, the Cape Verde region, located downwind over the eastern Atlantic, shows generally lower AOD levels. Notably, its AOD peaks slightly lag behind those in the source region, and its variation trends closely mirror the source's dynamics (e.g., concurrent rises on January 15 and 27). This spatiotemporal coherence clearly illustrates the process of dust emitted from the western Sahara being transported westward over the ocean by the prevailing easterly trade winds, consistent with the typical winter dust transport pathway in this region.

This case study demonstrates that the seamless AOD product generated in this study is effective for monitoring and analyzing typical atmospheric pollution processes. It is particularly suitable for aerosol events with distinct spatiotemporal variability and regional transport features, such as dust storms and biomass burning, thereby providing a reliable data foundation for regional environmental monitoring and research on climate effects.

V. Conclusion and Future Work

This study addresses the prevalent issues of high missing rates and insufficient accuracy in current satellite AOD products by proposing and validating a global AOD reconstruction framework that integrates multisource satellite products via OI fusion and seamless imputation using a transformer model. The main conclusions are as follows.

1) By fusing eight typical AOD satellite products from MODIS and VIIRS, the complementary advantages of multisource data were leveraged through the OI method, significantly improving data coverage and consistency. Validation against AERONET site data showed that the fused product increased the correlation coefficient by approximately 0.05–0.20, reduced RMSE by about 0.02–0.04, and increased the fraction of samples within the EE envelope to over 83% compared to individual source products, demonstrating its robust applicability on a global scale.

2) Building upon the fused product, the transformer-based imputation model effectively recovered missing regions using a triple self-attention mechanism across time, space, and meteorological variables. Experimental results showed that the imputed product achieved 100% spatial coverage and outperformed the MERRA-2 reanalysis product in accuracy, with R^2 exceeding 0.700 and RMSE reduced to 0.073, indicating superior stability and reliability.

3) In capturing typical pollution processes, the imputed results effectively reflected the accumulation and dissipation of pollution peaks, identified cross-regional transport features, and captured local dynamics, demonstrating the potential of the proposed method for studying spatiotemporal aerosol variations at both global and regional scales.

This study offers significant innovations. First, it achieves seamless global coverage of AOD products. By optimizing model architecture and training strategies, the proposed method extends high-accuracy reconstruction from regional to global scales, filling data gaps over historically underserved regions such as high-latitude areas, deserts, and tropical rainforests. Moreover, this work is the first to apply the transformer architecture to spatiotemporal AOD imputation tasks, introducing triple attention fusion across time, space, and auxiliary variables to adaptively integrate multidimensional features, thereby achieving organic synergy between heterogeneous data sources.

In summary, this study accomplishes high-accuracy and spatiotemporally continuous reconstruction of global AOD data, demonstrating the effectiveness of combining multisource fusion and deep learning-based imputation. It provides reliable data support for climate model evaluation and environmental research.

In this study, to harmonize satellite overpass times, we used data from three afternoon satellites with consistent overpass times. With the advancement of geostationary satellites, such as Himawari and GOES, incorporating high-quality geostationary satellite data products could further enhance global data supplementation. Therefore, future work may integrate such data to achieve more diversified product fusion. Due to computational constraints and data availability, the model in this study was trained and evaluated using global data from a single month. The one-month temporal span of the experimental data is insufficient to capture seasonal variations in aerosol loading. In future work, the model will be tested on datasets with longer temporal coverage to validate its temporal generalization capability, including seasonal performance and interannual changes. For the model itself, further improving the accuracy of imputation products and enhancing training efficiency remain important research directions. In addition, embedding physical mechanisms such as atmospheric transport models and boundary layer processes into the imputation framework could improve the physical consistency and interpretability of the results. These aspects warrant in-depth investigation in the future.

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organizations that provided the diverse research data essential for this work.

Specifically, the MODIS and Visible Infrared Imaging Radiometer Suite (VIIRS) aerosol optical depth (AOD) products were obtained from the Atmosphere Archive and Distribution System (LAADS). MERRA-2 AOD data can be accessed through http://disk.sci.gsfc.nasa.gov/mdisc/, while aerosol robotic network (AERONET) data were retrieved from the official AERONET webpage at http://aeronet.gsfc.nasa.gov/. In addition, ERA5 data were obtained from https://cds.climate.copernicus.eu/datasets

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